## Research activity

**Bounding an integral with rectangles — 35 min.**

We want to bound $I = \int_0^1 x^2 \, \mathrm{d}x$. For an integer $n \geq 1$, we split $[0,1]$ into $n$ intervals of equal length. Let $L_n$ be the sum of the areas of the rectangles whose heights are the values of $x^2$ at the left endpoints, and $U_n$ the sum obtained at the right endpoints.

1. For $n = 4$, draw the two families of rectangles, then calculate $L_4$ and $U_4$.
2. Justify that $L_4 \leq I \leq U_4$.
3. Write a Python function with argument $n$ that returns $L_n$ and $U_n$. Find $U_n - L_n$ algebraically.
4. Find the smallest integer $n$ for which this bound has a width of at most $0.01$.
5. Calculate the exact value of $I$ using an antiderivative and check that it lies within the bound obtained for $n = 4$.